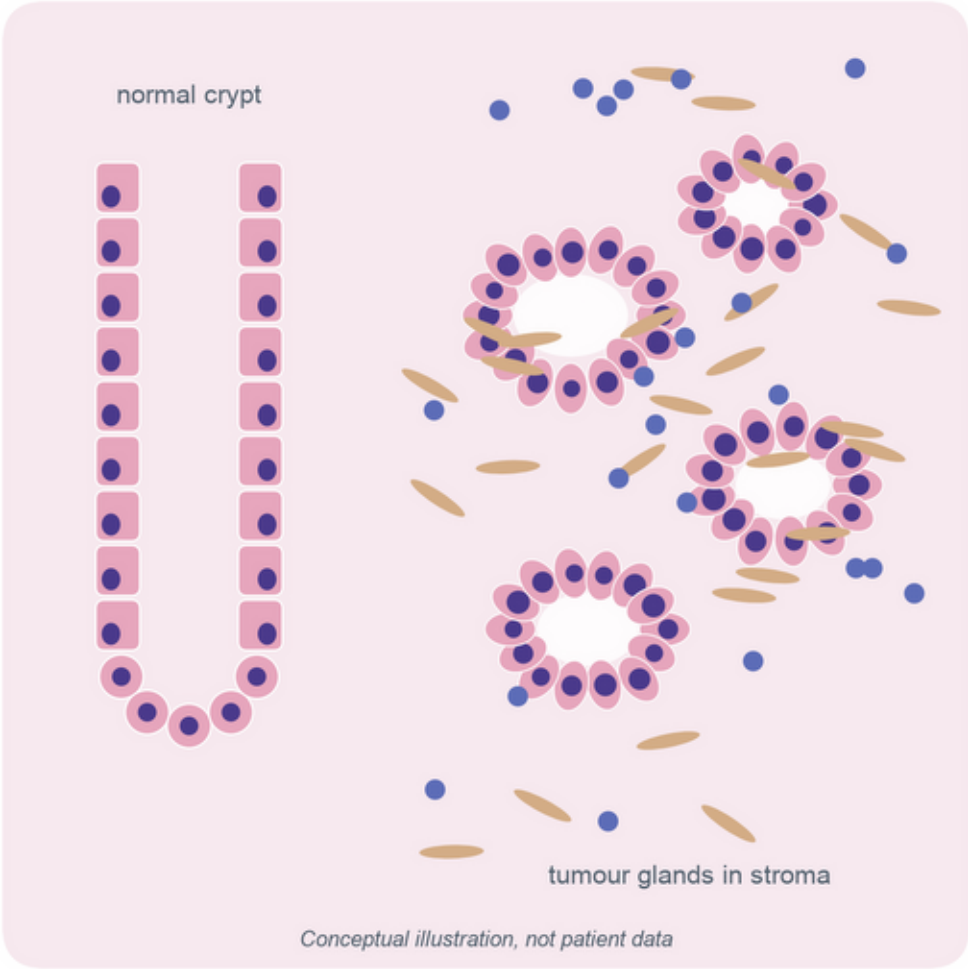
GSE39582 RECURRENCE STUDY

Do tumour pathway-expression scores improve colon-cancer recurrence prediction?

An optimism-corrected test of four Hallmark pathway scores in 452 stage II–III colon cancers (GSE39582), a four-cohort replication, and the pathway-audit tool



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MS in Bioinformatics, Northeastern University

November 2026

Illustration: original conceptual drawing, not patient data.

The clinical problem and the research question

1

The clinical problem

After surgery for stage II–III colon cancer, some patients recur and others do not. Stage alone sorts that risk imperfectly.

2

The question

Do four predefined tumour pathway-expression scores add recurrence-discrimination information beyond clinical variables?

3

The trap

A larger model usually fits its own patients better. That optimism makes signatures look more predictive than they are.

4

The approach

Fix the features before outcomes, compare clinical vs clinical + pathways, and measure the gain after removing optimism, with its uncertainty.

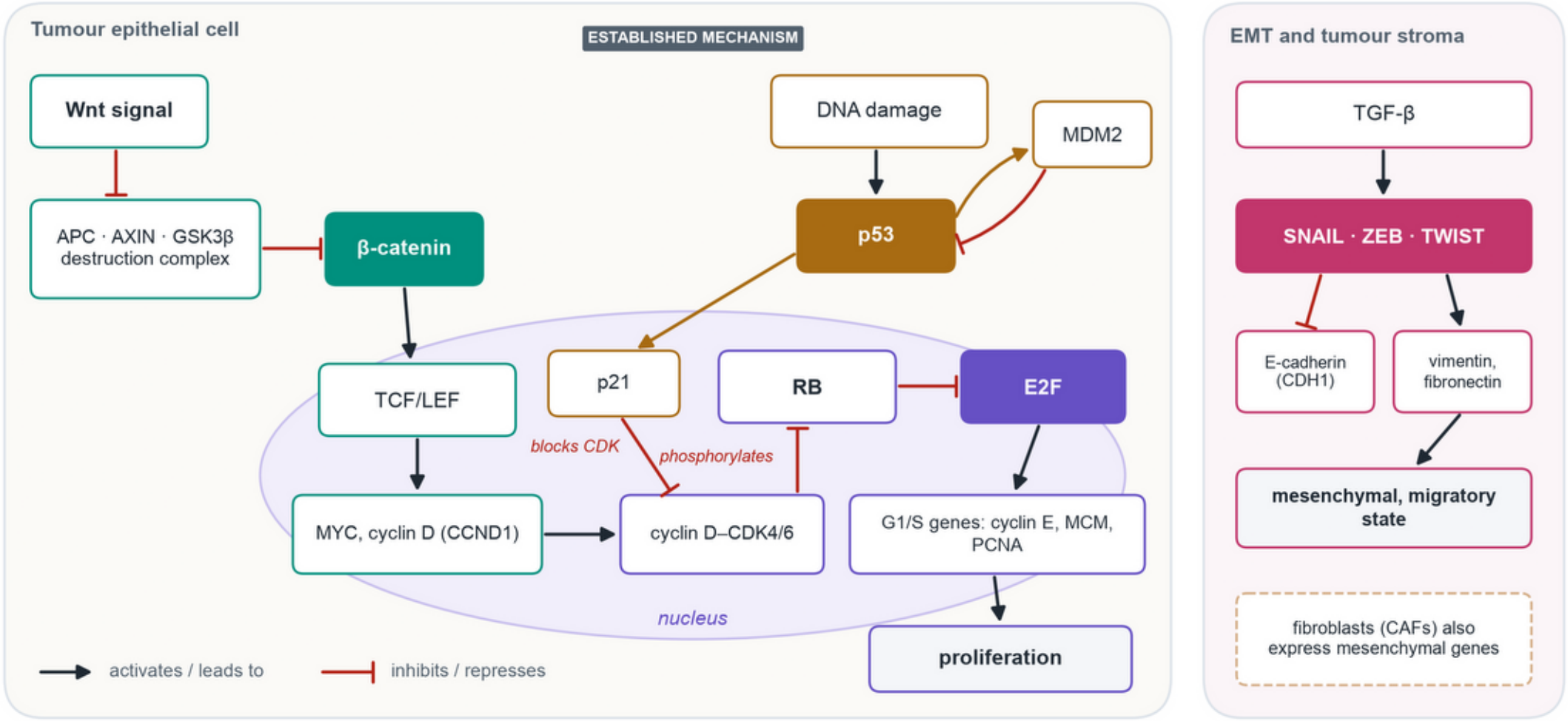
452	136	4	54,675
patients, stage II–III	recurrences	Hallmark pathway scores	probe sets, GEO GSE39582

Endpoint: cause-specific hazard of recurrence (not absolute risk). Validation: internal only.

Cohort counts: reports/figures/final/figure_values.json. Question: docs/PHASE6B_MODELING_PROTOCOL_DRAFT.md.

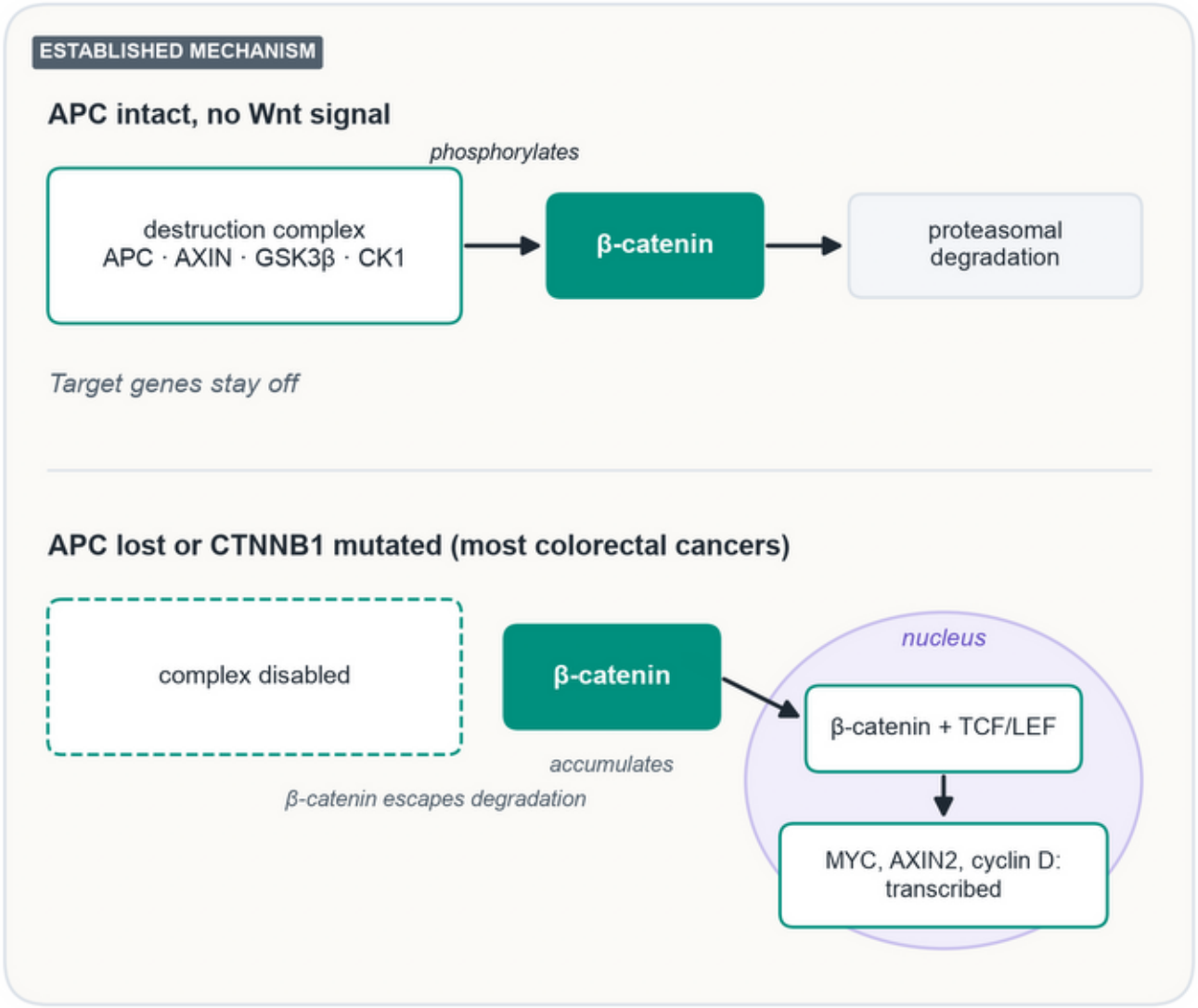
Four biological axes behind the pathway scores

Measured here: one bulk mRNA score per axis (EMT 198, E2F 195, Wnt 41, p53 194 genes). Pathway activity itself was not measured.



Established mechanisms (textbook background); gene sets: MSigDB Hallmark (Liberzon 2015).

Wnt/β-catenin: β-catenin stability controls TCF target genes



CRC LITERATURE

Wnt-pathway genes are genomically altered in 93 % of colorectal cancers; APC or CTNNB1 in about 80 % (TCGA 2012). Marked WNT/MYC activation characterises CMS2 (Guinney 2015). Genomic alteration is not measured activation.

MEASURED IN THIS STUDY

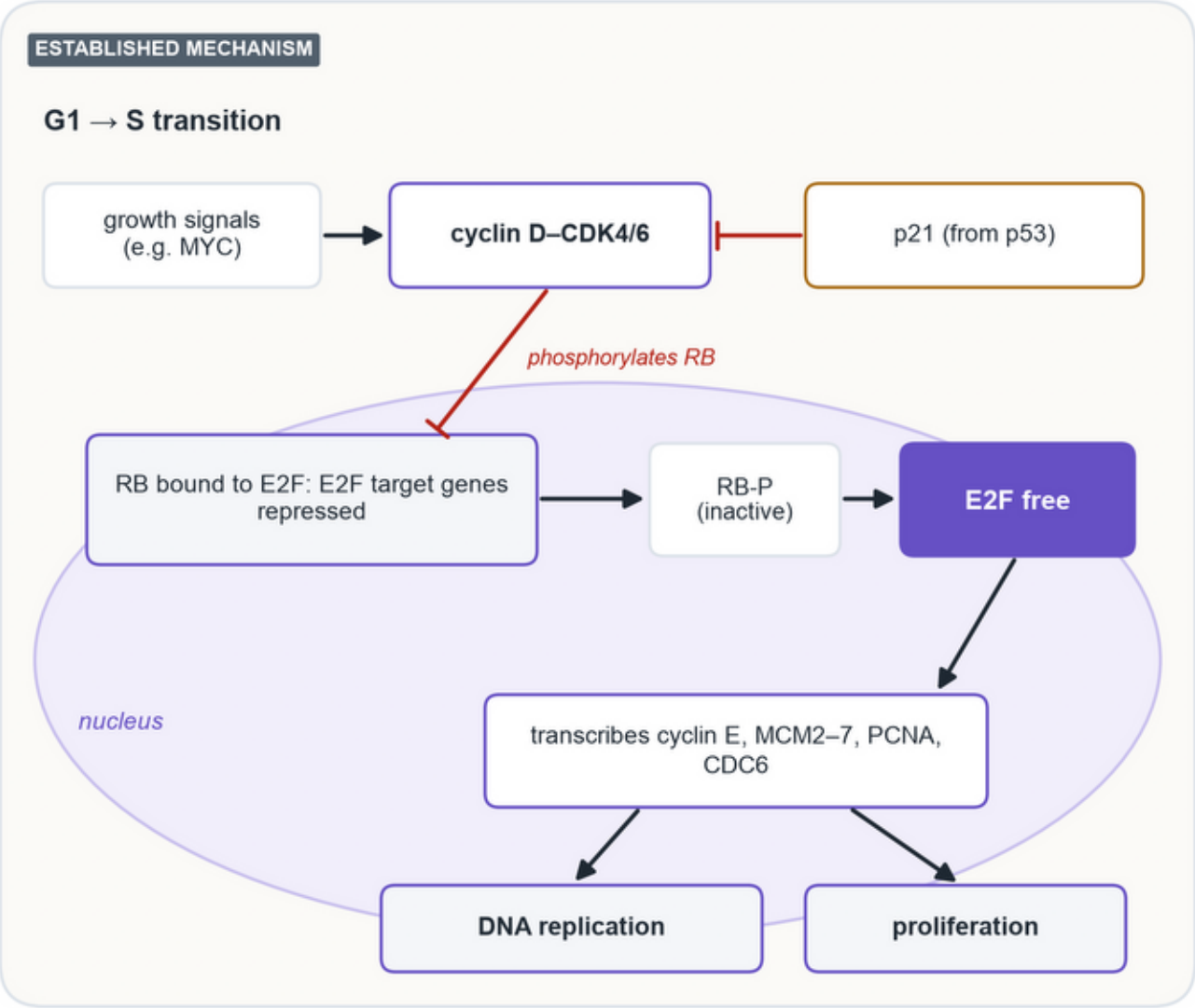
Mean standardised mRNA of 41 target genes. Primary HR 1.016 (0.906–1.138); S-7 1.011 (0.879–1.162). Within-set coherence 0.041 (loosely co-expressed). 1 sign change when single batches are omitted. Post-hoc (same data): lower in BRAF-mutant (-0.74 SD) and CIMP+ (-0.63 SD) tumours, one serrated axis.

NOT TESTED / HYPOTHESIS

Pathway activation (no nuclear β-catenin or APC data). No explanation is offered for the near-zero estimate.

Literature: TCGA 2012 (R6); Guinney 2015 (R1). Measured: phase9 execution and S-7 results; PHASE8C §3.1. Post-hoc: reports/exploratory/x1.

E2F: RB phosphorylation releases the G1/S programme



CRC LITERATURE

Evidence on proliferation and outcome is mixed: lower proliferation-signature expression in relapsed primaries (Anjomshoaa 2009), but high Ki-67 associated with poorer survival in a meta-analysis (Luo 2019).

MEASURED IN THIS STUDY

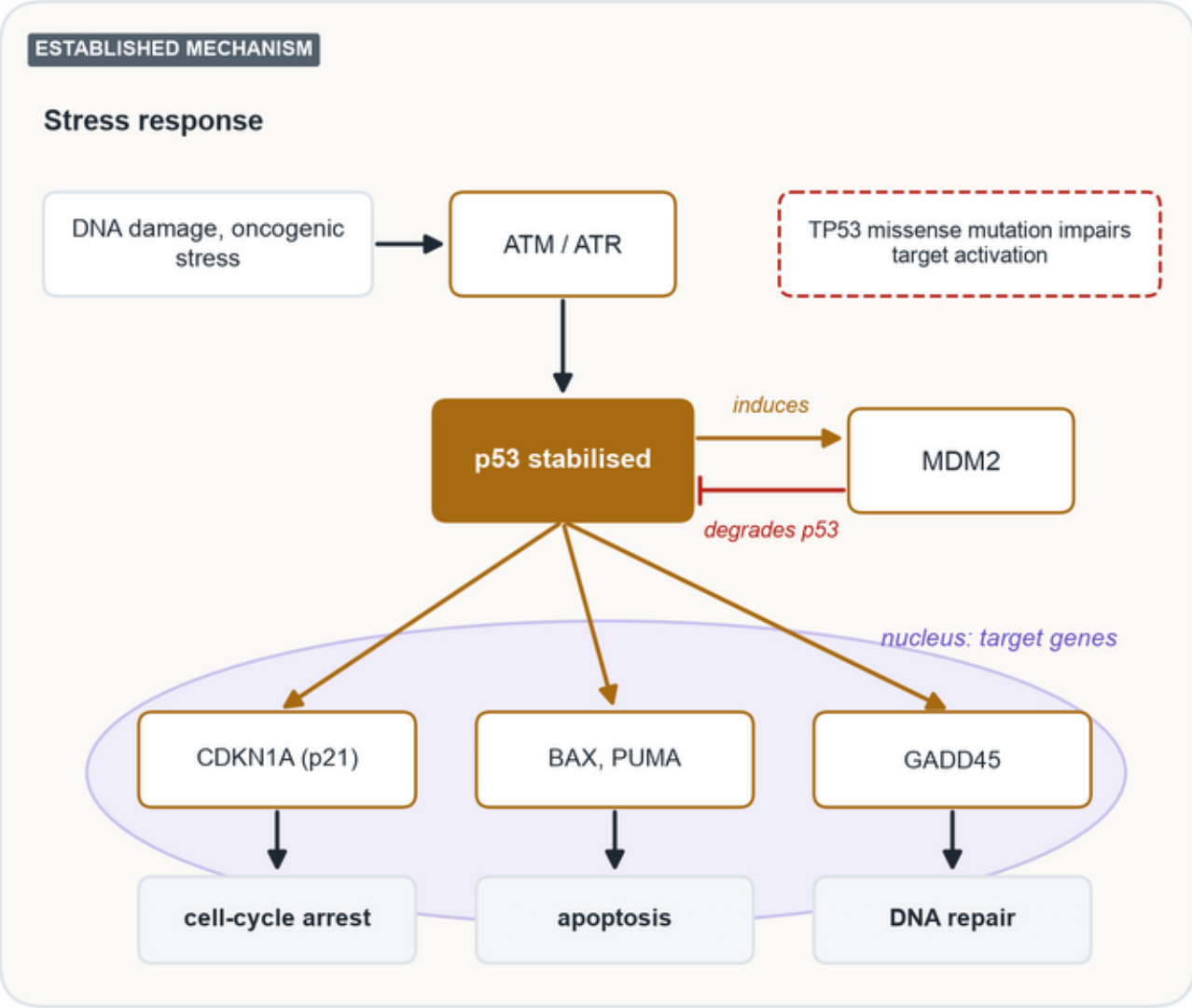
Mean standardised mRNA of 195 E2F target genes; coherence 0.304. Primary HR 0.906 (0.807–1.016); S-7 0.872 (0.758–1.003). Five-set model 0.862 (0.746–0.996): sensitivity only, coincided with added collinearity, not promoted.

NOT TESTED / HYPOTHESIS

Which cells are dividing, and whether proliferation causes or protects against recurrence. The direction is not established biology.

Literature: Anjomshoaa 2009 (R4); Luo 2019 (R5). Measured: phase9_execution_results.json; phase9_s7_results.json; PHASE8C §3.1.

p53: a stress checkpoint held in check by MDM2



CRC LITERATURE

TP53 is mutated in about 60 % of non-hypermutated and about 20 % of hypermutated colorectal cancers (TCGA 2012).

MEASURED IN THIS STUDY

Mean standardised mRNA of 194 p53 target genes. Primary HR 0.995 (0.882–1.123); S-7 0.993 (0.852–1.158). Strongest MMR association of the four scores (ε^2 0.1716, exploratory); 2 sign changes under batch omission. Post-hoc: lower in TP53-mutant tumours (-0.45 SD; also within pMMR), replicated in GSE41258 and TCGA-COAD (rank-biserial -0.22, -0.42).

NOT TESTED / HYPOTHESIS

p53 protein function and activity, and the mechanism of the MMR association. A mutation–expression link is not a function assay.

Literature: TCGA 2012 (R6). Measured: phase9 execution and S-7 results; phase8c/score_vs_annotation.csv. Post-hoc: reports/exploratory/x1, x4.

EMT: tumour-cell programme or stromal signal?

ESTABLISHED MECHANISM

Two biological sources of the same mesenchymal genes

1 · Tumour cells undergoing EMT



2 · Fibroblasts in the tumour stroma



CAFs express collagens, vimentin and fibronectin without any change in the tumour cells

one bulk EMT score (198 genes)

An elevated bulk EMT-associated score does not establish tumour-cell EMT.

CRC LITERATURE

The mesenchymal CMS4 subtype had worse relapse-free survival (Guinney 2015; GSE39582 contributed, so not independent). Mesenchymal-subtype transcripts in bulk tumours are mostly stromal (Calon 2015; Isella 2015).

MEASURED IN THIS STUDY

Mean standardised mRNA of 198 genes. Primary HR 1.070 (0.951–1.202), above 1 in every fit but never distinguishable from 1; S-7 1.084 (0.937–1.254). MMR association ε^2 0.0394 (exploratory). Post-hoc: tracks fibroblast markers with shared genes removed (ρ 0.94–0.98) in GSE39582 and three independent cohorts.

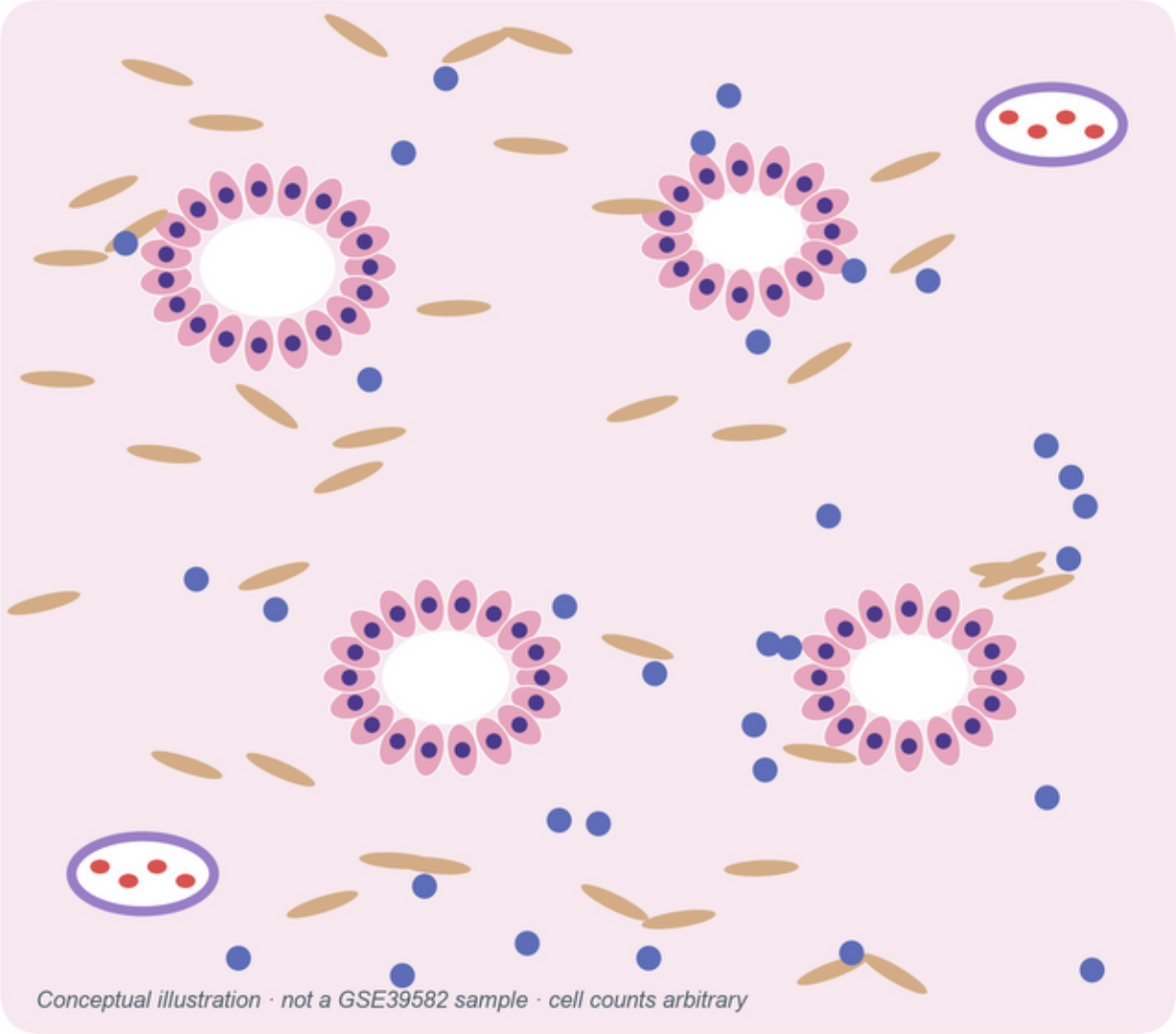
NOT TESTED / HYPOTHESIS

The cellular source. Composition was estimated computationally, not measured; bulk data cannot separate tumour-cell EMT from stroma.

Literature: Guinney 2015 (R1); Calon 2015 (R2); Isella 2015 (R3). Measured: phase9 results. Post-hoc: reports/exploratory/x5, x6.

WHAT WAS MEASURED

What a bulk tumour sample measures



Tumour epithelial cells

NOT SEPARATED · purity estimated computationally (ESTIMATE)

Fibroblasts and stroma

NOT MEASURED · abundance estimated (ESTIMATE, MCP-counter)

Immune cells

NOT MEASURED · abundance estimated (ESTIMATE, MCP-counter)

Endothelial cells and vessels

NOT MEASURED · abundance estimated (MCP-counter)

Mixed tissue
all cell types

→

One array
pooled mRNA, one value per gene

→

One score
mean of member-gene z-scores

Post-hoc estimates from the same arrays: EMT score ρ 0.95 with estimated fibroblast content. Estimates, not measured fractions.

Conceptual illustration. Estimates: reports/exploratory/x5 (ESTIMATE v1.0.13; MCP-counter v1.2.0), post-hoc.

GSE39582: from 585 arrays to 452 analysed patients

- 585 Affymetrix HG-U133 Plus 2.0 arrays, deposited as RMA + ComBat (Marisa 2013).
- 452 stage II–III patients: 136 recurrences, 316 censored.
- Endpoint: cause-specific hazard of recurrence; 46 deaths without recurrence are censored (informative).
- Recurrence-time unit undocumented: rank-based measures only.
- CIT partition (99 patients, 29 events) is internal, never external validation.

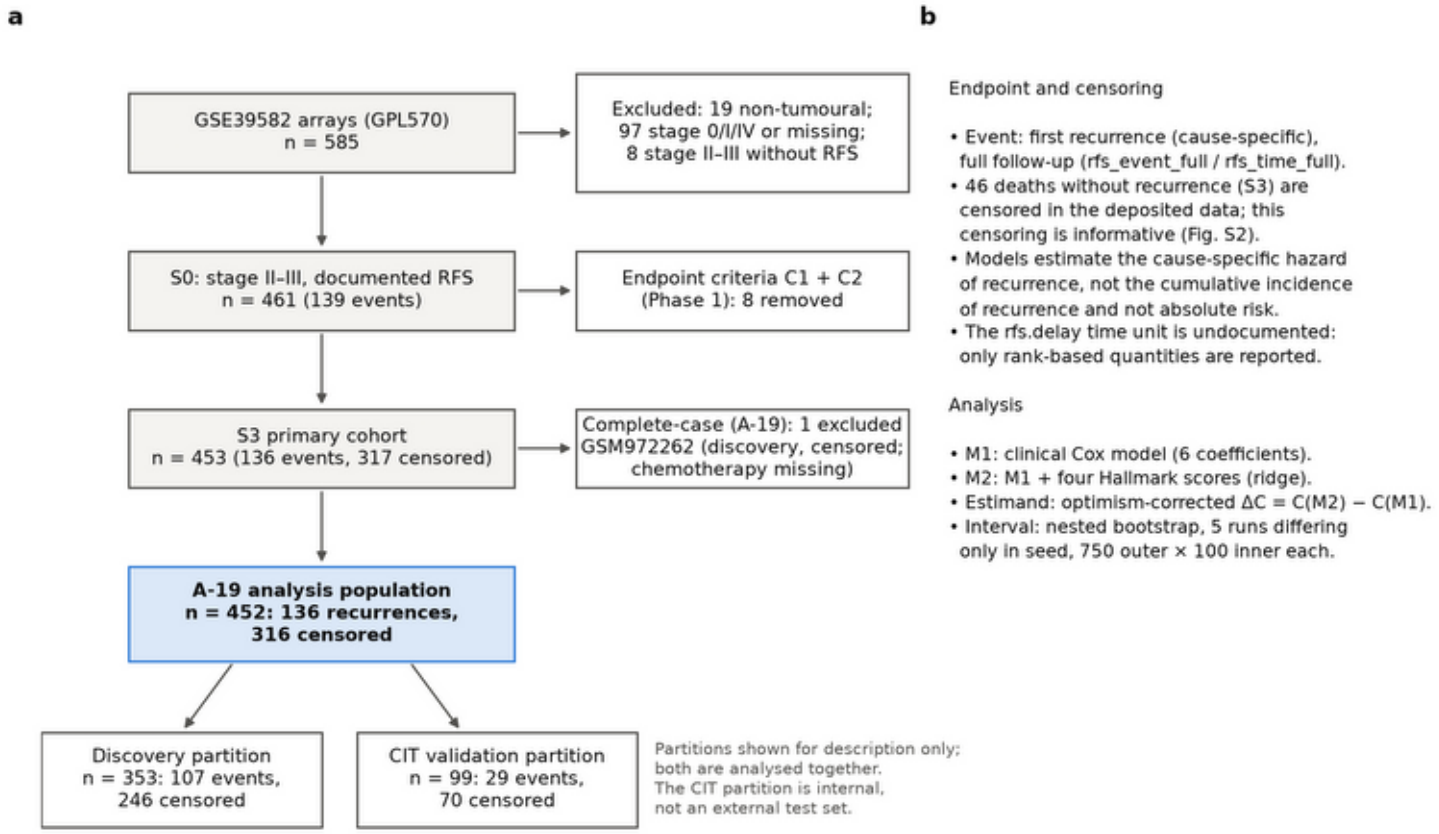


Figure 1 (reports/figures/final/Fig1_study_design_cohort.png). Censoring profile: appendix A2.

From 54,675 probe sets to four pathway scores

54,675 Affymetrix probe sets — HG-U133 Plus 2.0; deposited RMA + ComBat values

43,163 map to one gene — 8,893 unmapped and 2,619 ambiguous probes excluded

21,815 unique genes — median of probes per gene (10,751 genes have several)

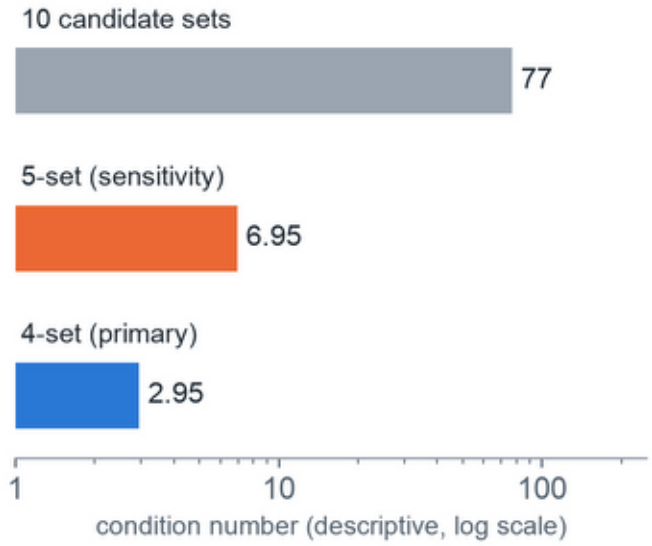
607 pathway-member genes — EMT 198 · E2F 195 · Wnt 41 · p53 194

4 scores per patient — standardised inside each training fold

Bar widths follow log feature count (last bar enlarged for legibility).

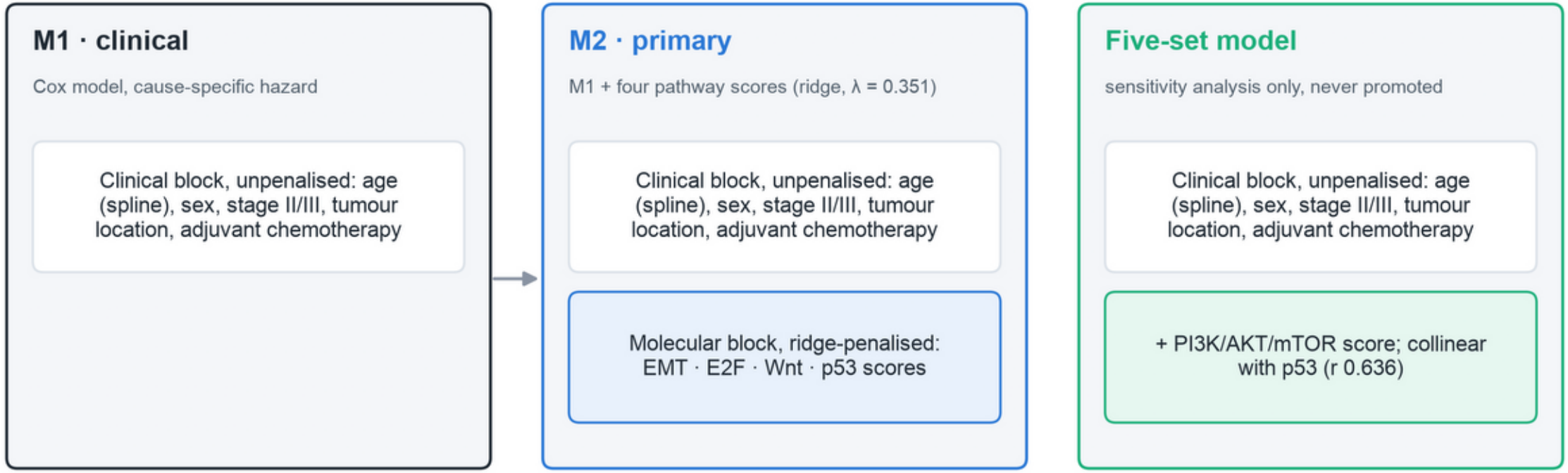
Outcome-blind panel selection

Ten candidate Hallmark sets formed correlation clusters (six on one mesenchymal axis, r 0.60–0.91 with EMT). One representative per cluster was chosen without outcome data.



figure_values.json (fig2.probes); config/pathways/m2_panel_4set.json and m2_panel_5set.json; PHASE6C §5.

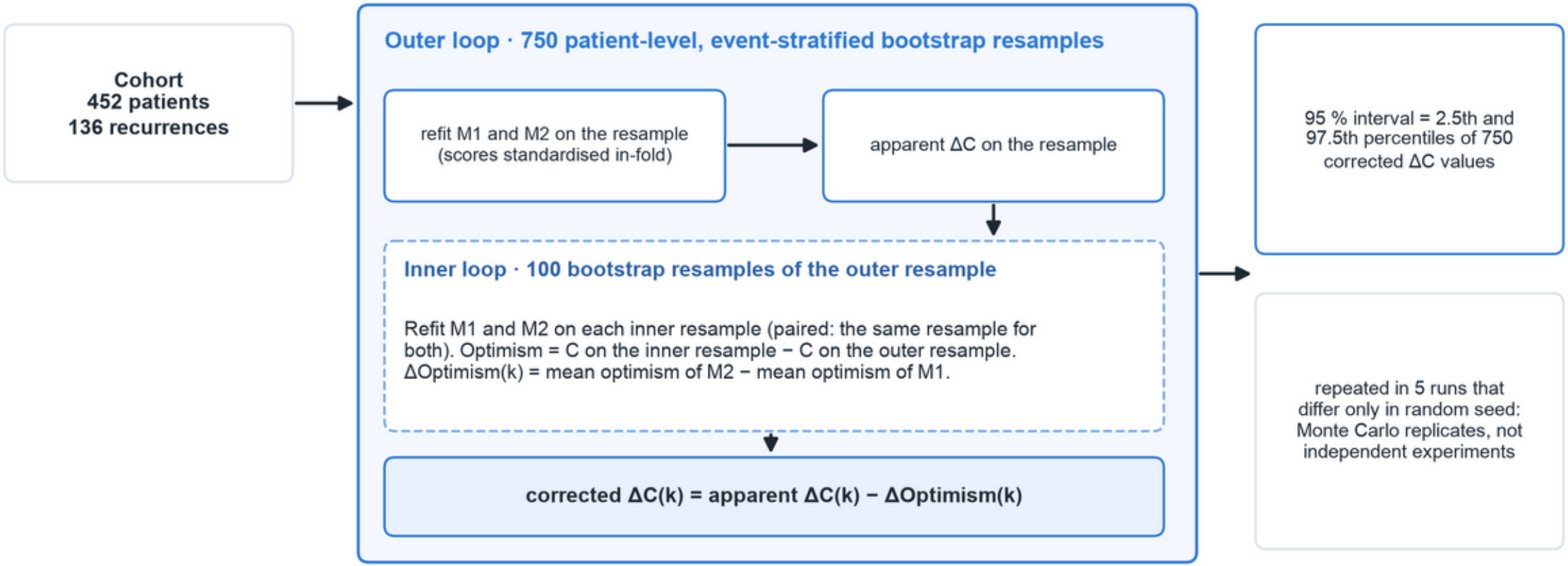
Model architecture: clinical benchmark and molecular extension



- M1 is exactly M2 with the pathway coefficients set to zero: the comparison isolates the molecular block.
- The clinical block is unpenalised, which keeps the comparison scale-equivariant; shrinking only the molecular block makes the gain in C conservative.
- λ is tuned by five-fold, event-stratified cross-validation inside each training set; M1 is never re-tuned.

docs/PHASE6B_MODELING_PROTOCOL_DRAFT.md §3; λ from phase9_execution_results.json.

Validation: optimism-corrected nested bootstrap



Precision rule (pre-specified): largest endpoint Monte Carlo SE ≤ 5 % of interval width; achieved 4.2 % after topping up from 500 to 750 outer resamples.

Point estimate: full-data corrected ΔC = +0.004316. Reporting was gated on a frozen verifier (PASS 60/60); a separate re-implementation recomputed the reported quantities from the checkpoints.

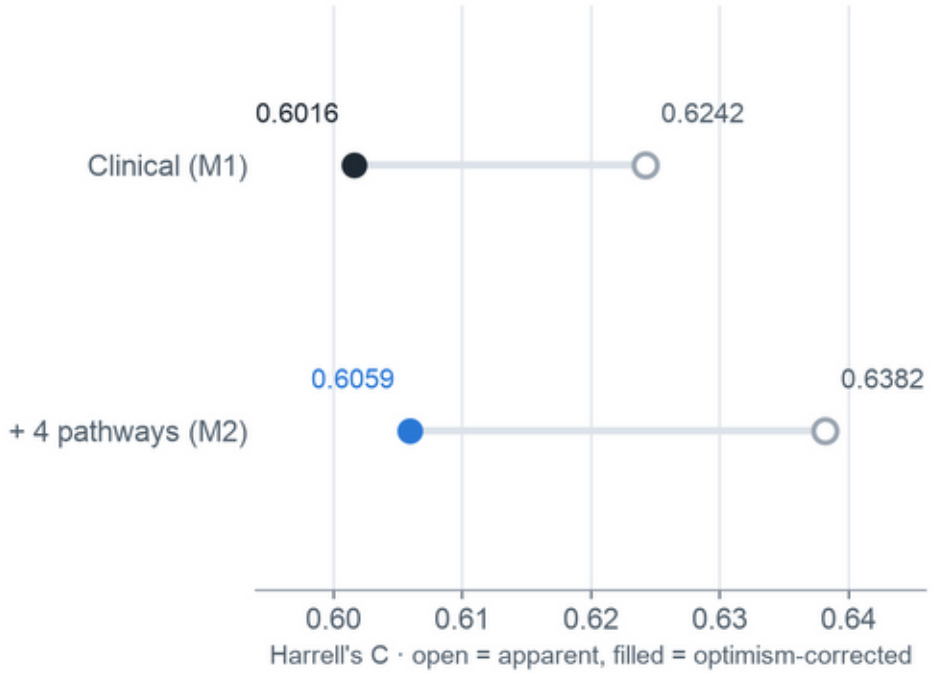
docs/PHASE9_DELTAC_INTERVAL_PLAN.md §3; figure_values.json (fig8, figS1); verifier output.

CENTRAL RESULT

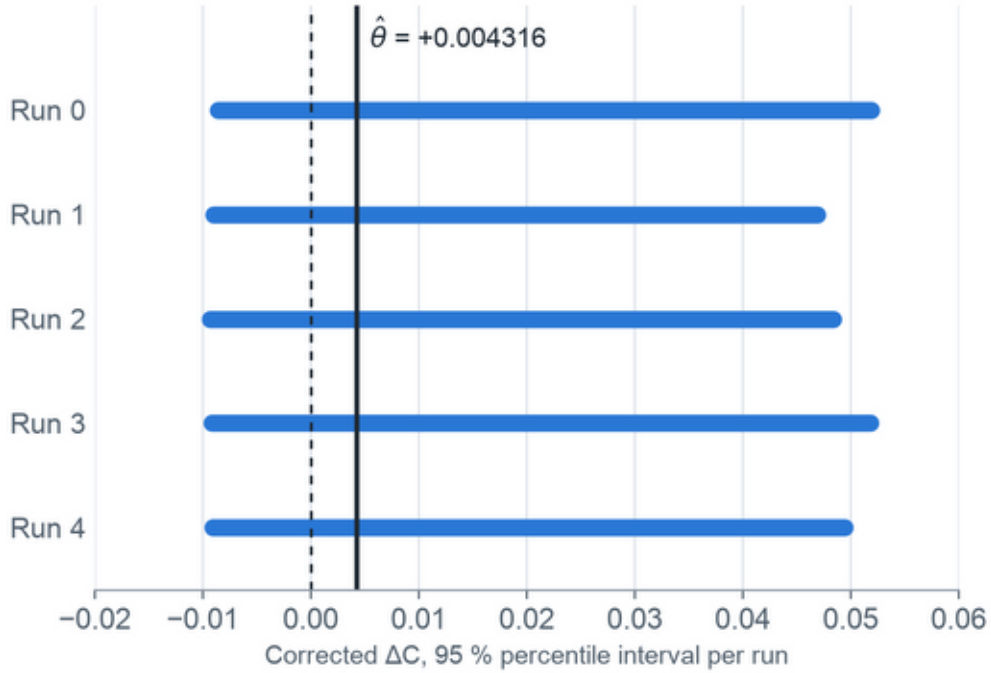
Central result: no demonstrated gain in discrimination

Optimism-corrected gain in C: +0.004316. All five 95 % intervals include zero, so no reproducible improvement was demonstrated.

1 · Most of the apparent gain was optimism



2 · All five corrected intervals include zero

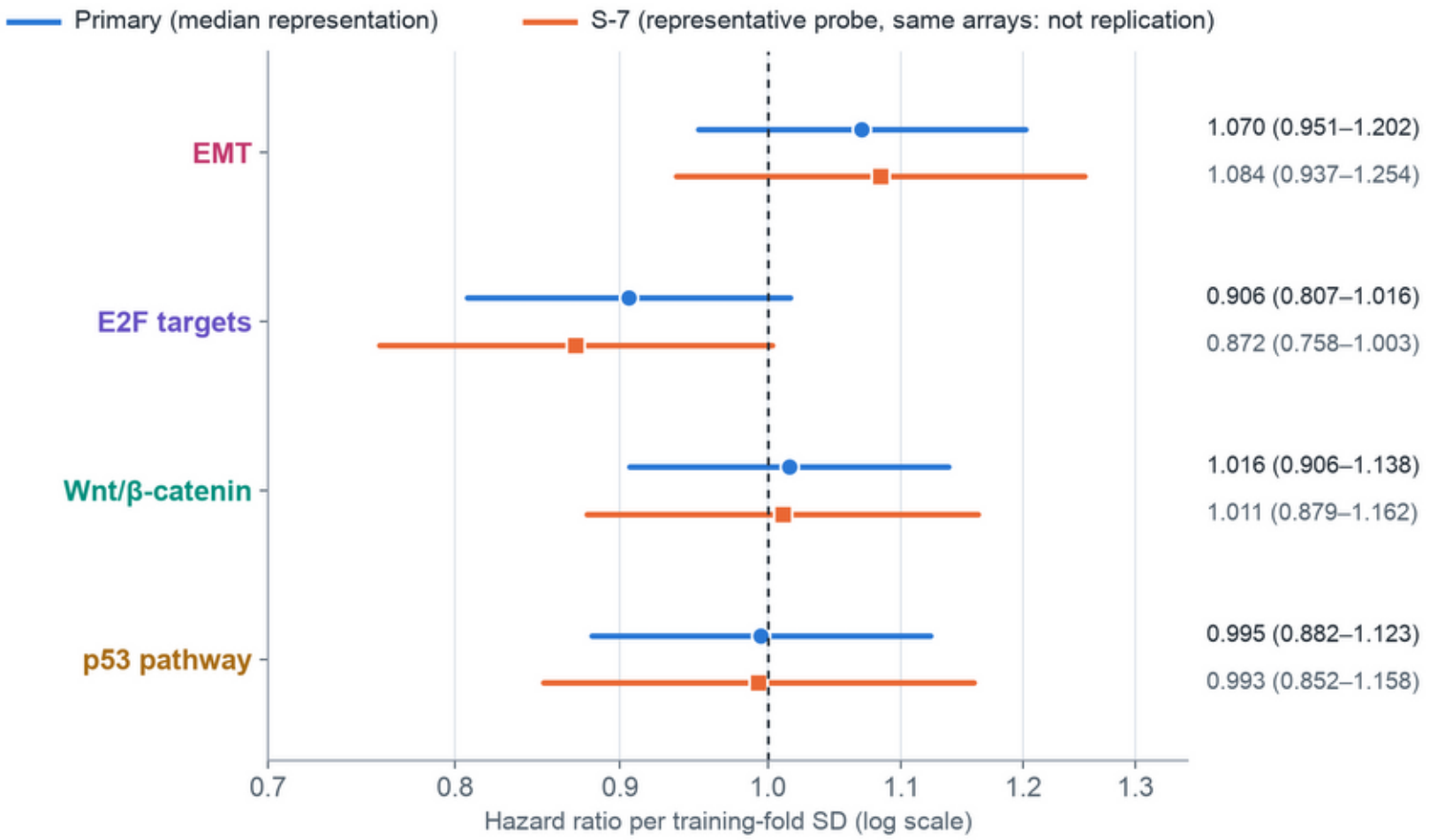


Apparent ΔC +0.0140 → corrected ΔC +0.004316 · intervals ≈ −0.009 to +0.05 · calibration slopes 0.833 (M1), 0.862 (M2): residual overfitting

The study did not demonstrate a reproducible improvement in recurrence discrimination from adding the four pathway scores.

figure_values.json (fig7, fig9); reports/phase9/corrected_rerun_report/corrected_interval_results.json.

Pathway hazard ratios: every primary interval includes 1



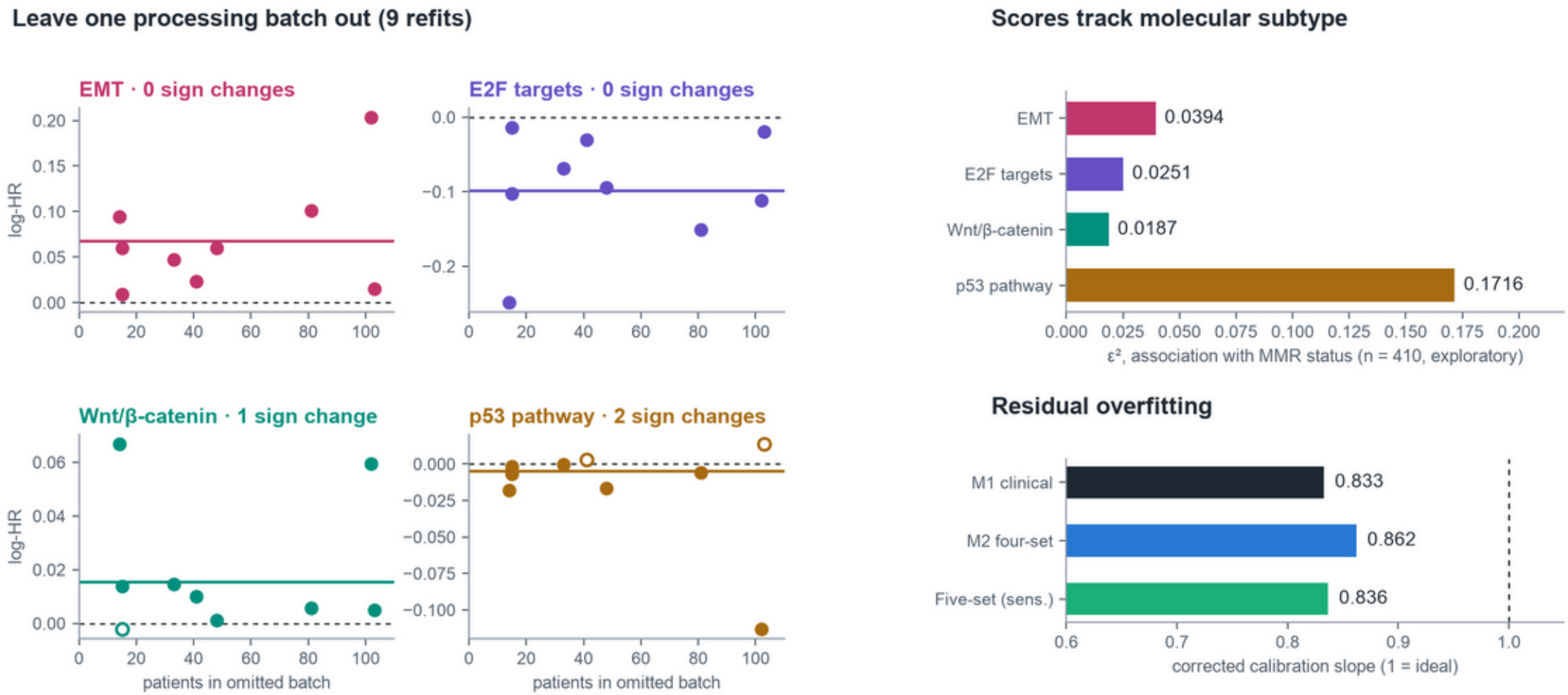
Sensitivity only

Five-set model E2F HR 0.862 (0.746–0.996). Excludes 1 only after PI3K was added (collinear with p53); fixed in advance as sensitivity, not promoted.

No primary-model interval excludes 1. Wald intervals are approximate under ridge penalisation.

phase9_execution_results.json (M2 primary, M2 sensitivity); phase9_s7_results.json.

Robustness: batch sensitivity, subtype association, calibration



Omitted batches hold 14–103 patients, so the spread mixes batch identity with sample size. Open circle = sign differs from the all-patient estimate (solid line). Batch effects were characterised, not resolved.

phase9_execution_results.json (batch_sensitivity); score_vs_annotation.csv; figure_values.json (fig9).

CONTRIBUTIONS

What this project established

1 A reusable pathway-audit framework

TESTED SOFTWARE

Python package and command-line tool that checks what a gene-set score measures, in bulk tumour data and in single cells, before it is interpreted.

Data-quality gate → Gene drivers, coherence → Tissue composition → Molecular associations → Random-set null → Survival screen

13/13 package tests pass

CI passes on Python 3.10 and 3.12 (GitHub Actions)

4/4 planted truths recovered in the demo

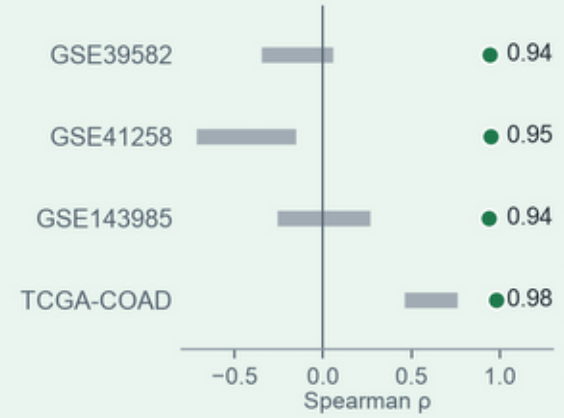
5 + 1 bulk + single-cell case-study cohorts

Version 0.2.0. Docker image written, not yet built.

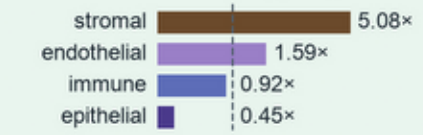
2 Replicated: EMT score tracks stroma

BULK ×4 · SINGLE-CELL ×3

Bulk: gene-disjoint ρ 0.94–0.98 in four cohorts, each above its own random-gene-set band (grey).



Single cells (discovery): EMT-set transcript share vs matched random genes; dashed = 1×



Tumour-cell EMT programme not detected in single cells (not ruled out: rare or partial states).

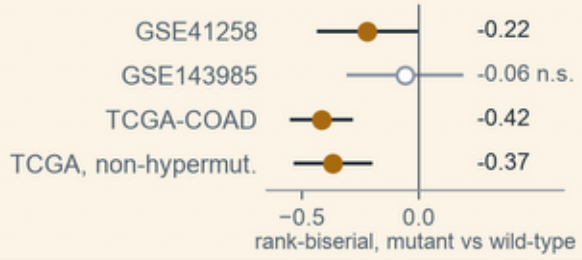
3 Molecular anchoring: p53 score tracks TP53

REPLICATED · 2 OF 3 EXTERNAL

Lower in TP53-mutant tumours; replicated in GSE41258 and TCGA-COAD, and after excluding hypermutated tumours.

An mRNA-score association, not a measure of p53 protein activity or pathway function.

Not established: E2F–TP53 and p53–MSI (not replicated); TCGA BRAF links (explained by hypermutation).



4 Rigorous predictive evaluation

PRE-SPECIFIED

Corrected ΔC +0.004316; all five 95 % intervals include zero.

No reproducible gain in recurrence discrimination. Does not rule out a biological role.



What remains unresolved

Tumour-cell EMT
not detected in single cells; partial states need spatial or protein data

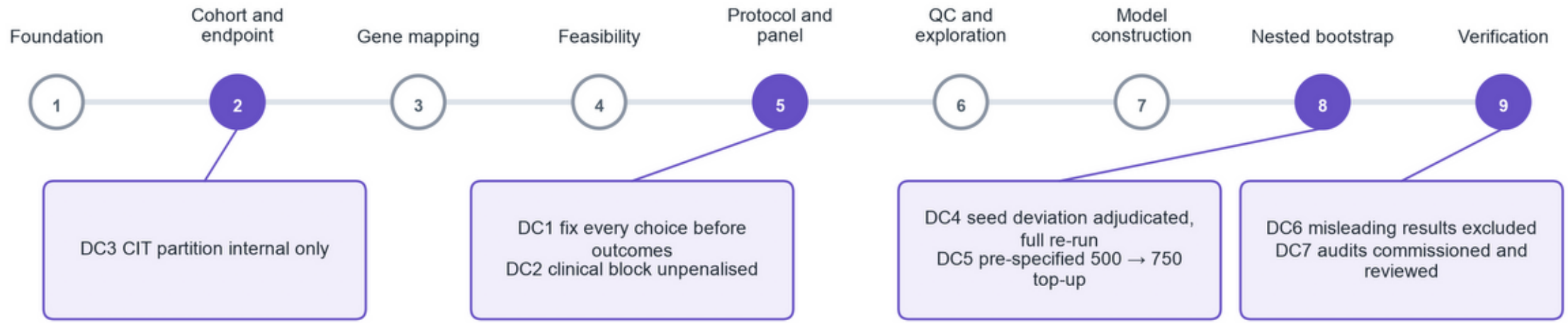
Direct pathway activation
needs functional or protein-level measurement

External predictive validation
no eligible cohort: 0 of 14 assessed; max 124 vs ≈337 recurrences needed

Recurrence-time unit
undocumented; provisionally months

tools/pathway-audit/; docs/RESEARCH_EXTENSION_REGISTER.md + TCGA addendum. All limitations: docs/SCIENTIFIC_LIMITATION_RESOLUTION_REGISTER.md.

Decisions and audit trail



60 / 60

frozen verifier checks passed before reporting

2.2×10^{-16}

maximum difference in the independent re-computation (from checkpoints; not a model refit)

138

reported quantities recomputed independently

1,097

automated tests passing

My contribution: scientific direction and methodological decisions, each recorded in the project's decision records (DC1–DC7).

docs/OWNER_DECISION_REGISTER.md; docs/PHASE9_CORRECTED_FINAL_REPORT.md §13–§15.

From finding to solution: replication and the pathway-audit tool

1 · Replicated finding: the EMT score reads stroma



2 · Delivered as a tool: pathway-audit

1 Data-quality gate	control gene pairs must co-express
2 Gene drivers	coherence, leave-one-out, stable modules
3 Composition	stromal / immune estimates, gene-disjoint test
4 Genotype effects	robust to, or explained by, MMR and composition
5 Survival screen	optimism-corrected gain over clinical model
6 Single-cell sources	which cell types supply a set's transcripts

Case studies: five bulk cohorts and one single-cell cohort

- EMT composition-dominated in all 4 usable cohorts
- GSE103479 stopped at the quality gate
- BRAF → p53 score explained by MMR and composition (+0.89 → +0.03 SD)
- TCGA: BRAF effects explained by hypermutation; TP53 → p53 score robust
- Survival screen on GSE39582: corrected ΔC +0.0048 (frozen +0.0043)
- Single cells (GSE132465): EMT transcripts mainly stromal (5.1×), epithelial 0.45×

Post-hoc registered extensions, outcome-free; tool tested (13/13; CI passing) incl. ESTIMATE match to 1e-14. Composition values are estimates, not measured fractions.

docs/RESEARCH_EXTENSION_REGISTER.md and TCGA addendum (registered before analysis); tools/pathway-audit/; reports/pathway_audit_case_study/.

Conclusion and the next scientific questions

How the project answered its question

Asked — Do four pathway scores improve recurrence prediction after surgery?

Found — No reproducible gain: corrected ΔC +0.004316, every interval includes zero.

Asked next — What do these scores measure in bulk tumour tissue?

Found — EMT score tracks stroma in 4 bulk and 3 single-cell cohorts; p53 score tracks TP53 status.

Delivered — pathway-audit, so these checks can be run before a score is interpreted.

Take-home

Audit bulk pathway scores for tissue composition, genotype confounding and data quality before reading them as biology or adding them to prognostic models.

Next scientific questions

NOT TESTED / HYPOTHESIS

Q1 Are there rare or partial tumour-cell EMT states that single-cell RNA missed?

How: Spatial transcriptomics or multiplexed protein imaging (e.g. E-cadherin, vimentin) at the invasive front.

Why: Single cells placed EMT-set transcripts mainly in stroma, with no coherent tumour-cell programme; localised states need spatial data.

Q2 Are the Wnt, E2F, p53 and EMT pathways actually active?

How: Functional or protein-level readouts: reporter assays, phospho-protein and target-protein measurement.

Why: Turns expression associations into evidence about pathway function.

Q3 Does any score add recurrence information in a new cohort?

How: A pre-specified test in an independent stage II–III cohort with ≈337 recurrences and documented follow-up time.

Why: None of the 14 public cohorts assessed was eligible (largest: 124 recurrences).

docs/SCIENTIFIC_INTERPRETATION_AND_CONCLUSION.md; docs/FINAL_RESEARCH_CONTRIBUTION.md; docs/EXTERNAL_COHORT_ELIGIBILITY_AUDIT.md.

APPENDIX A1

References and figure sources

References

- R1 Guinney J et al. Nat Med 2015;21:1350–1356. doi:10.1038/nm.3967
- R2 Calon A et al. Nat Genet 2015;47:320–329. doi:10.1038/ng.3225
- R3 Isella C et al. Nat Genet 2015;47:312–319. doi:10.1038/ng.3224
- R4 Anjomshoaa A et al. Br J Cancer 2009;101:822–828. doi:10.1038/sj.bjc.6605229
- R5 Luo ZW et al. BMC Cancer 2019;19:123. doi:10.1186/s12885-019-5324-y
- R6 TCGA Network. Nature 2012;487:330–337. doi:10.1038/nature11252
- R7 Liberzon A et al. Cell Syst 2015;1:417–425. doi:10.1016/j.cels.2015.12.004
- R8 Marisa L et al. PLoS Med 2013;10:e1001453. doi:10.1371/journal.pmed.1001453

Figure sources

- Slide visuals: portfolio/diagrams/slides/ (original vector drawings by portfolio/src/make_slide_visuals.py, numbers read from frozen and registered post-hoc outputs).
- Replication data: GEO GSE41258, GSE143985, GSE103479; TCGA-COAD via UCSC Xena. Tool: tools/pathway-audit/.
- Figures 1, 2 and S2: reports/figures/final/ (generated from frozen outputs by scripts/49).
- Mechanism panels are textbook background, labelled as established mechanism; no external images are used.
- Full results: docs/SCIENTIFIC_RESULTS_AND_FIGURES.md and docs/PHASE9_CORRECTED_FINAL_REPORT.md.

References verified in docs/BIOLOGICAL_EVIDENCE_AND_CITATION_AUDIT.md.

APPENDIX A2

Quality control and informative censoring

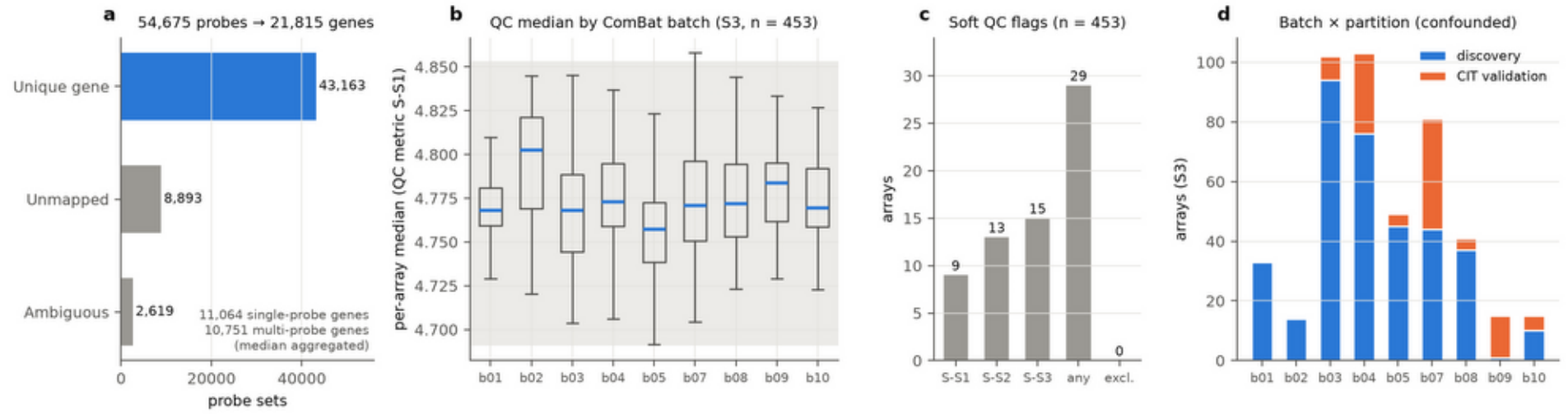


Figure 2. Probe mapping, per-array QC (29 soft flags, none excluded) and batch confounded with partition.

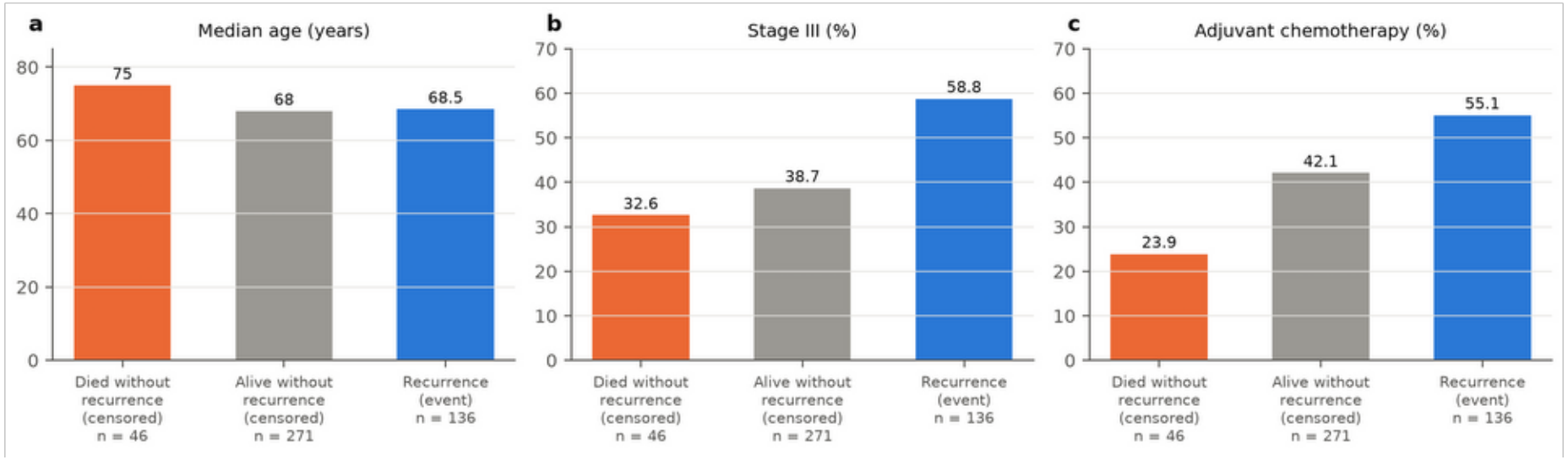


Figure S2. The 46 patients censored at death without recurrence differ from the rest: censoring is informative.

Figures 2 and S2 (reports/figures/final/).